

SQL for Security Telemetry Analysis & Endpoint Compliance Segmentation

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Environment / Engine: MariaDB / MySQL RDBMS

Target Schemas: log_in_attempts (Security Ingress Telemetry), employees (Enterprise Asset Directory)

Primary Focus: Forensic Incident Triage, Brute-Force Anomaly Isolation, Targeted Patch Rollout Engineering

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1.0 Executive Summary & Technical Context

Modern Security Operations Centers (SOCs) and IT infrastructure teams process millions of unstructured and semi-structured event logs daily. While native SIEM environments index high-level alerts, direct querying of operational Relational Database Management Systems (RDBMS) via Structured Query Language (SQL) is indispensable for low-latency incident forensics, correlation of boundary anomalies, and granular data segmentation during critical vulnerability remediation cycles.

This project details the construction, optimization, and operational deployment of structured SQL queries across two core security initiatives:

1. **Threat Detection & Ingress Triage:** Isolating after-hours brute-force access vectors, establishing temporal threat boundaries around a known breach window, and identifying geolocation ingress anomalies.
2. **Endpoint Compliance & Patch Deployment:** Programmatically isolating endpoint cohorts based on geographic site parameters, organizational business units, and universal negative filters to drive targeted vulnerability remediation

2.0 Ingress Telemetry Forensics & Incident Investigation

The investigative phase evaluated authentication metrics within the *log_in_attempts* schema, containing connection telemetry: *event_id*, *username*, *login_date*, *login_time*, *country*, *ip_address*, and *binary connection status (success)*.

2.1 Isolating Off-Hours Authentication Failures (Brute-Force & Credential Triage)

Identify potential off-hours brute-force, password-spraying, or automated credential-stuffing campaigns by isolating failed authentication attempts occurring outside standard operating hours (after 18:00)

SQL Execution Syntax:

```
SELECT *  
  
FROM log_in_attempts  
  
WHERE login_time > '18:00' AND success = FALSE;
```

This query targets all operational telemetry within the `log_in_attempts` dataset. The **WHERE** clause applies a conditional boundary using the logical **AND** Boolean operator to force the evaluation engine to satisfy both parameters simultaneously. The first constraint isolates entries with a value greater than standard business operations (`login_time > '18:00'`). The second filter utilizes a boolean evaluation gate to isolate only failed connections (`success = FALSE` or 0).

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

```
19 rows in set (0.018 sec)
```

Figure 2.1(lists of off-hours brutal force attack)

2.2 Temporal Boundary Analysis Across an Active Exploitation Window

Threat intelligence identified a confirmed perimeter intrusion active on 2022-05-09. To detect initial network probing, reconnaissance, or persistent access, the audit window was expanded to include the 24 hours preceding the incident date (2022-05-08 through 2022-05-09).

SQL Execution Syntax:

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

This query scans the log_in_attempts table. Because a connection cannot exist on two calendar dates simultaneously, using an AND operator would yield zero results. Instead, the logical OR operator was implemented to instruct the database engine to retrieve rows matching either the date of discovery (2022-05-09) or the preceding observation window (2022-05-08).

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
52	amazon	2022-05-08	11:51:28	CAN	192.168.122.188	1

Figure 2.2(lists of off-hours brutal force attack within the last 24 hours)

2.3 Cross-Border Geolocation Anomaly Auditing

Threat analysis confirmed that the recent wave of malicious login attempts did not originate from authorized infrastructure nodes located within Mexico. To eliminate this benign geographic traffic and isolate international ingress vectors, a cross-border filtering query was generated.

SQL Execution Syntax:

```
SELECT *
FROM Log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

This query reads from the log_in_attempts table. To account for data entry variance where Mexico is stored as either MEX or MEXICO, the LIKE operator was combined with the

percentage sign (%) wildcard. The MEX% pattern instructs the database engine to look for any string starting with those three specific characters. By prepending the logical NOT operator, the database flips the criteria, securely outputting all global records *except* those originating within the designated country boundaries.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0

Figure 2.3(Filtering login attempts outside Mexico)

3.0 Enterprise Asset Segmentation for Patch Orchestration

The remediation phase shifted focus to the corporate asset repository (employees), containing *employee_id*, *device_id*, *username*, *department*, and *office*.

3.1: Physical Facility & Departmental Cohort Isolation

The endpoint security team issued a high-priority advisory to deploy updates to machines assigned to the Marketing division. Crucially, the configuration update applies specifically to assets located within the East building complex.

To retrieve the exact assets target list for this deployment, the following data segmentation query was run:

```

SELECT *

FROM employees

WHERE department = 'Marketing' AND office LIKE 'East%';

```

This query shifts context to parse the internal directory database (employees). The logical Boolean **AND** operator restricts the output rows to assets matching two distinct conditions. The first clause isolates users assigned to the specified organizational division (*department* = 'Marketing'). The second clause utilizes the pattern-matching operator *LIKE* along with the wildcard trailing operator ('East%') to search the office column. This ensures that only endpoints residing in rooms such as East-170, East-195, or East-267 are compiled into the patch target sheet.

```

MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Marketing' AND office LIKE 'East%';

```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

```

7 rows in set (0.001 sec)

```

Figure 3.1 (Lists of assets in the East building)

3.2 Logical Grouping for Multi-Department Software Patches

A critical security configuration patch required immediate simultaneous deployment across all machines in the **Finance** and **Sales** business units. A consolidated target directory was required to initiate the automated deployment cycle.

SQL Execution Syntax:

```

SELECT *

FROM employees

WHERE department = 'Finance' OR department = 'Sales';

```

Because individual employees are bound to a single functional business unit row, the **OR** statement allows the query handler to execute an aggregation, extracting rows where either

the Finance parameters match (*department = 'Finance'*) or the Sales criteria are met (*department = 'Sales'*).

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	ireckley	Finance	Central-295

Figure 3.2 (Lists of employees in either finance or sales department)

3.3 Universal Exclusion Targeting (Out-of-Scope Cohort Generation)

The internal IT Infrastructure department successfully deployed updates across all machines within the **Information Technology** department. An audit query was required to generate a target list of all remaining endpoints across all other departments to ensure universal organizational coverage.

SQL Execution Syntax:

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

This query searches rows within the employees corporate matrix table. The logical Boolean **NOT** operator within the syntax scans the schema and filters by displaying the data for all employees except those whose **department = 'Information Technology'**.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrhah	Marketing	West-246
1028	c603d749e374	astrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375

Figure 3.3 (Filtered employees in all department except information technology)

4.0 Executive Summary & Operational Impact

The strategic construction and optimization of targeted relational database queries successfully converted high-volume corporate logging repositories into actionable security telemetry. By deploying precise boolean logic (**AND**, **OR**, **NOT**) alongside pattern-matching wildcards (**LIKE '%'**), critical forensic triage was conducted to isolate post-business-hours authentication spikes and cross-border ingress anomalies. Concurrently, programmatic data segmentation enabled the precise cohorting of enterprise endpoints across facilities

and business units, accelerating targeted security configuration rollouts and patch compliance. These database manipulation capabilities establish a scalable technical baseline for automating infrastructure auditing, supporting real-time Security Operations Center (SOC) triage, and mitigating attack surfaces across modern enterprise environments.